Exploring Company Stock Behavior Based on News Article Headline Sentiment Using FinBERT

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1. Introduction

The stock market has proven to be affected by the sentiment of financial news sources due to the influence news sources have over investors [1]. Due to the vast amount of articles and blogs that get posted in the current day and age, it has been difficult to accurately quantify the effect that financial news has on company stock prices. The recent technological advancements to large language models and transformers has made it easier to collect and analyze this data, but has proven to be ineffective in accurately predicting whether articles have a positive or negative sentiment toward a company [2]. Additionally, LLMs cannot adjust to the volatile nature of the stock market because of the lack of market patterns [3]. To get around this problem, some studies have manually annotated articles to ensure the models being used are not misrepresenting articles [4]. However, this can be a time consuming and expensive tradeoff.

Despite the limitations of LLMs, one has come to the forefront when it comes to examining the sentiment from financial texts. FinBERT is a variation of Google's BERT which has been trained on thousands of financial texts in order to capture the sentiment from these texts. While finBERT has achieved accuracies of nearly 90% [5] and proven to be the premier LLM for sentiment analysis of financial texts, it is unclear if finBERT's can be used to predict company stock price. This study aims to determine if news article headlines scored by finBERT can be used as an effective predictor for company stock price.

2. Literature Review

One study uses LLMs such as finBERT and and Llama 3.1 to gather the sentiment from news articles about the top 30 companies from the NASDAQ to capture the effect articles have on company stock price [5]. The study then employed three different investment strategies, and found that using the sentiment found, only a long-term strategy was effective. Another study wants to make advancements in sentiment detection with the use of LLMs, in this case finBERT [6]. The researchers collect sentiment from news articles using finBERT and predict stock prices which are compared to simulated market results to look for areas of improvement. Similarly to both of these studies, I also plan to use finBERT to gather sentiment analysis from news articles. However, I plan to focus solely on the article headlines, while these studies use the entire article.

The stock market as a whole moves according to financial sentiment whether that be from news articles or social media, and it is important to quantify the sentiment to predict stock market behavior. One study aims to do this using the content from financial news articles and Twitter posts in conjunction with word2vec and BERT sentiment indexes for sentiment analysis and convolutional neural networks for forecasting [7]. This looks into how sentiment analysis affects the stock market as a whole, while I'll be looking at how sentiment analysis affects stocks on a company-level. Other areas of sentiment analysis include social media posts, as they also have an effect on stock market prices and should be quantified so investors know the extent of this effect. A study utilizes parts-of-speech tagging for sentiment analysis of twitter posts along with linear regression and deep neural networks for stock price prediction [8]. This study uses similar techniques to the ones used in the current study for both sentiment analysis and prediction, but does not focus on news articles.

Sentiment analysis may have a greater impact on companies of a specific sector, and is looked into by a study that researches sentiment analysis among energy companies [9]. It uses finBERT for sentiment analysis and long-short-term memory models for prediction among seven energy companies in the New York Stock Exchange over a ten year period. This study uses a centralized approach, looking at a few specific companies over a long period of time, while my objective is to look at a broad range of companies over a short period of time. It is important to explore sentiment analysis from social media posts using differing collection methods. For instance, one study used graph neural networks with natural language processing methods to collect sentiment analysis and long-short-term memory models for prediction [10]. My objective differs as I am exploring sentiment analysis from news articles rather than social media posts.

3. Methodology

Data Collection & Cleaning

This study aims to quantify the sentiment of news article headlines on company stock price. By using BeautifulSoup for web scraping and ThreadPoolExecutor to scrape multiple titles simultaneously, every article title on the first page of Google's news section was collected for every day of 2025 when each Fortune 500 company was searched. For instance, on June 16, 2025 an article titled “Tesla leads U.S. sales of EVs lower in April, marking the first annual drop in over a year” is a result when “Tesla” is the search value in Google, thus is being used as data for this study. To gather the company stock price, this study will be using Yahoo Finance. Specifically, the closing price will be gathered from the yf python library for every day in 2025 for each company being analyzed.

Fortune 500 companies were chosen due to their high reputation, making them more likely to have plenty of news articles written about them. That being said, most of these companies have days where there are no articles written about. For example, when “HF Sinclair” is searched, there are no results from September 14th to 17th, meaning no articles were published about this company during that time frame. To ensure that only companies with a sufficient amount of articles written about them are being used, only companies with at least 330 days of 5 or more articles written about them were used. This left only 55 publicly traded companies for the study.

To quantify the sentiment of news article titles, this study will be using finBERT, an LLM pre-trained on financial texts that assigns text probabilities based on how likely it is to be positive, negative, or neutral [11]. The main metric used in this study is acquired by applying this LLM to every news article title, then taking the difference of the average positive and average negative scores for each day among each company. This metric will be referred to as “sentiment score” in the study. For days where a company had less than 5 articles written about them, the most recent previous sentiment score was used.

Stock Performance Analysis & Evaluation

To determine if the sentiment gathered from news article headlines using finBERT is effective at predicting company stock price, a baseline first has to be established. For this study the baseline will be a long short-term memory (LSTM) model using a rolling seven day window as the input. LSTMs are a type of neural network architecture that forgets information that is no longer useful to avoid an overloading of information while retaining useful information to be used for prediction [12]. LSTMs are great for sequential data and will be a good fit for evaluating stock prices over time. Root mean squared error (RMSE) will be used to evaluate model performance.

RMSE measures the distance between predicted values and actual values, while penalizing larger differences and using the same units as the target variable. Time series errors typically follow a normal distribution, which is the distribution best suited for RMSE [13].

There will be three base models in this study. The baseline, which will use the daily percent change of stock price of the previous seven days to predict the daily percent change of the stock price on the next day. The second base model will use the sentiment scores from the previous seven days to predict the daily percent change of the stock price on the next day. The last base model will use both the percent change of stock price and the sentiment scores from the past seven days to predict the stock price change on the next day. The objective is to determine if the second and third models yield better results than the baseline model.

These base models will be applied to each company individually. Additionally, each model will progress through differing amounts of lag; one day, two days, three days, and one week. Lag meaning the number of days after the seven days of stock price change, and/or sentiment score. In total, there will be 660 models spanning across the companies and lag amounts.

4. Results

Table 1 showcases the average root mean squared error (RMSE) of the LSTM models based on the days of lag and predictor variables. Price_only refers to models that used only the stock price change over the past seven days as a predictor and is the baseline model for this study. Sentiment_only refers to models that used only the sentiment scores over the past seven days as a predictor, and combined refers to models that used both as predictors. Sentiment_vs_price displays the average differences between the sentiment_only models and the price_only models, whereas combined_vs_price displays the average differences between the combined models and the price_only models.

Table 2 has the same variables as table 1, but displays the median RMSE instead of the average RMSE. Table 3 shows the percentage of models for each type of lag where the difference in RMSE is positive, indicating the altered model performed better than the baseline model.

Table 1:

Days of Lag	price_only	combined	sentiment_only	sentiment_vs_price	combined_vs_price
1	1.80868209%	2.04685343%	1.86703977%	-0.05835768%	-0.23817134%
2	1.83000808%	2.04196629%	1.85535205%	-0.02534397%	-0.21195821%
3	1.83641022%	2.07641866%	1.88388024%	-0.04747002%	-0.24000844%
7	1.85077764%	2.06610321%	1.84739877%	0.00337887%	-0.21532557%

Table 2:

Days of Lag	price_only	combined	sentiment_only	sentiment_vs_price	combined_vs_price
1	1.62674302%	1.87053869%	1.67884510%	-0.03324790%	-0.18856661%
2	1.66520218%	1.87247541%	1.67784142%	-0.01334598%	-0.12434833%
3	1.62871541%	1.85630505%	1.73167934%	-0.01670639%	-0.14188085%
7	1.61705302%	1.85492329%	1.66099358%	-0.04381512%	-0.20085794%

Table 3:

Days of Lag	sentiment_vs_price	combined_vs_price
1	32.73%	21.82%
2	47.27%	23.64%
3	49.09%	29.09%
7	43.64%	21.82%

The model using only stock price change as a predictor had the lowest RMSE by mean and median for all amounts of lag with the one exception being seven days of lag where the model using only sentiment score as a predictor had a slightly lower mean. This implies that using sentiment score as a predictor is worse for model performance. Conversely, the model using both stock price change, and sentiment score, as predictors had the highest mean and median RMSE across all amounts of lag. This suggests that the combined model performed the worst across all amounts of lag. Overall, using finBERT's sentiment score from news article headlines to predict company stock price movement is worse than baseline LSTM models using previous company stock price movement as a predictor.

To further support this claim, we can compare RMSE values from different models on a company basis. From table 3, we can see that the highest percentage of models with a positive difference in RMSE is 49.09%. This means that the majority of altered models performed worse by RMSE than the baseline model.

5. Conclusion

The goal of this study was to determine if the sentiment analysis gathered using the finBERT LLM could be used as a predictor for company stock price. Ultimately, the findings suggest that sentiment analysis from finBERT actually hurts model performance, and LSTMs performed better using just previous stock movement as a predictor. However, this should not be a conclusive result that finBERT's sentiment analysis is definitively ineffective in predicting company stock price. This study only looked at article titles and a seven day window of previous sentiment score and stock price movement. Further research should examine the effects of the continent of articles as well as different time periods of previous stock price change and sentiment.

6. References

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